

Module 1: Introduction to Data Mining and Big Data

Instructional Hours: 6

Module Overview

In this module you will be introduced to the world of data mining and big data, focusing on real-world applications, particularly in industries like agriculture, healthcare, and finance in Kenya. The goal is to help you understand how data-driven decisions are changing industries globally and locally. You will explore how data mining assists in decision-making, pattern discovery, and predictive analytics, critical in sectors such as agriculture, finance, healthcare, and government.

Learning Outcomes

At the end of this module, you should be able to:

1. Define data mining and big data.
2. Differentiate between structured, semi-structured, and unstructured data.
3. Choose appropriate data mining techniques for real-world problems.
4. Review big data analysis techniques.

Instructional Materials & Learning Activities

Data mining and big data are intertwined areas shaping the modern data-driven world. In Kenya, data is invaluable in agriculture, healthcare, finance, and governance. Proficiency in these areas empowers you to solve problems, build predictive models, and make informed decisions.

1. What is Data Mining?

- **Definition:** The process of discovering patterns and insights from large datasets.
- **Key Steps:** Data cleaning, transformation, pattern discovery (e.g., clustering, classification), evaluation, interpretation.
- **Example:** Predicting student performance from historical exam scores, attendance, and study habits.

2. Key Steps in Data Mining

1. Data Collection (databases, sensors, social media, surveys).
2. Data Cleaning (handling missing values, correcting errors).
3. Data Integration & Transformation (merging sources, feature engineering).
4. Pattern Discovery (classification, clustering, association rules).
5. Evaluation & Interpretation (accuracy, relevance, actionability).

3. Why Data Mining Matters in Kenya

- **Agriculture:** Optimize crop yields and farming practices.
- **Healthcare:** Predict outbreaks, identify risk factors.
- **Finance:** Detect fraud in mobile money (e.g., M-Pesa).
- **Education:** Tailor programs, track skills gaps.

4. What is Big Data?

- **Definition:** Extremely large and complex datasets exceeding traditional system capabilities.
- **Characteristics (4Vs):** Volume, Velocity, Variety, Veracity.
- **Example:** Real-time data from M-Pesa, social media, e-commerce.

5. Big Data in the Kenyan Landscape

- **Finance:** Mobile money transactions.
- **Transport:** Route mapping, traffic analysis.
- **Governance:** Census, healthcare, education records.

6. Relationship Between Data Mining and Big Data

- Big Data provides **infrastructure** (distributed storage, processing).
- Data Mining provides **algorithms** (classification, clustering).
- Together: Generate actionable insights.

7. Big Data Tools and Frameworks

1. Apache Hadoop (HDFS, MapReduce, YARN).
2. Apache Spark (in-memory processing, MLlib, GraphX, Spark SQL).
3. NoSQL Databases (MongoDB, Cassandra, HBase).
4. Cloud Platforms (AWS, Azure, GCP).
5. Workflow Tools (Airflow, NiFi, Luigi).
6. Visualization Tools (Tableau, Power BI, Grafana).

8. Data Mining Techniques

- **Classification:** Decision Trees, SVM, Neural Networks.
- **Regression:** Linear Regression, Random Forests.
- **Clustering:** K-means, DBSCAN.
- **Association Rules:** Apriori, FP-Growth.
- **Dimensionality Reduction:** PCA, t-SNE.

Assessments

1. Explain the interdependence between data mining and big data. [5 marks]
2. What is Hadoop, and why is it important in processing large datasets? [5 marks]

Assignments

- Compare two clustering algorithms (e.g., K-Means vs. DBSCAN) on a Kenyan dataset. Write a report on performance metrics and significance.

Quizzes

1. Which dimensionality reduction technique preserves variance? (A) PCA (B) K-Means (C) OLAP (D) Apriori [2 marks]
2. Which big data characteristic focuses on data reliability? A. Volume B. Velocity C. Variety D. Veracity
3. Which technique groups data without pre-labeled categories? A. Classification B. Clustering C. Regression D. Association
4. Apache Spark is preferred over MapReduce for: A. Storing files B. Spreadsheets C. In-memory ML tasks D. Transactional DBs
5. Which is true about association rule mining? A. Only for text B. Finds hidden patterns C. Only time-series D. For data cleaning
6. One major ethical concern in big data projects is: A. Interfaces B. Cost C. Privacy misuse D. Data collection difficulty

Mini Project

Acquire or simulate a dataset. Clean and explore it, possibly applying dimension reduction. Select one classification or clustering algorithm to analyze the data. [20 marks]

Reference List

1. Han, J., Pei, J., & Kamber, M. (2023). *Data Mining: Concepts and Techniques* (4th ed.). Morgan Kaufmann.
2. Leskovec, J., Rajaraman, A., & Ullman, J. D. (2020). *Mining of Massive Datasets* (3rd ed.). Cambridge.
3. Mayer-Schönberger, V., & Cukier, K. (2017). *Big Data: A Revolution*. Mariner Books.
4. Marz, N., & Warren, J. (2015). *Big Data: Principles and Best Practices*. Manning.
5. Russell, S. J., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson.

Case Study: Safaricom M-Pesa

Safaricom's M-Pesa processes millions of transactions daily. Fraud detection requires big data pipelines (HDFS, Spark) to process transaction logs in real time. Data mining algorithms (clustering, anomaly detection) help identify fraudulent patterns.

Questions:

1. Discuss possible algorithms for real-time fraud detection.
2. Primary purpose of data mining in this case: (B) Discover patterns and anomalies.
3. Which big data characteristic is most relevant? (C) Velocity.
4. Why prefer Spark over MapReduce? (C) In-memory iterative processing.
5. Best anomaly detection technique without labels? (C) Clustering.
6. Major ethical concern? (B) Privacy and misuse of financial data.

Online Discussion

What privacy and data handling laws are critical for financial institutions? Post your answer in the online forum.

Take-Home Reflection

What one key aspect of big data and data mining have you appreciated most? Record in your journal.

What's Next

Next, you will learn about **Data Warehousing**, acquiring knowledge and skills to store and manage data in repositories.